

REEFPRINT/KHANYA: ore-hardness prediction on the belt and sulphide characterisation in the laboratory.

An analogue-data proof of concept and staged validation plan for South African concentrators

Mintek–SCi Grad Hackathon 2026 · Problem 3: Computer vision for real-time mineralogical characterisation

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Abstract

South African concentrators set mill feed, grind and reagents before the laboratory can say what was in the ore. REEF-PRINT/KHANYA combines hyperspectral prediction of grinding hardness on the belt with microscope classification of sulphide minerals in the laboratory, and passes both to an operator who reviews every proposed change. On 146 public porphyry-ore composites the hardness model reached an out-of-fold R^2 of 0.479 (mean absolute error 1.05 kWh/t, against R^2 0.210 for colour alone), and a nominal one-sided 90% upper bound covered 93.2% of samples. A simplified feed-rate replay estimated a 1.7–1.9% throughput gain over a fixed design-feed policy; this is a simulation without a plant time series. On 12 public microscope sections that were reused during development, class IoU was 0.870 for pyrrhotite, 0.576 for chalcopyrite and 0.547 for pentlandite. The advisor gave definite advice on 4 sections, consistent with the same rules applied to expert masks, and abstained on 8; this is not independent evidence of decision accuracy or plant safety. Neither subsystem has been tested on South African ore. We therefore propose local, independently evaluated mineralogy and hardness studies first, and a belt shadow trial only if they pass. The financial scenarios show that small effects could repay a belt, but they need the host’s own production boundary, costs and measured process response before investment.

1 Introduction

As Team Sonar, working on Mintek’s Problem 3, our responsibility is to give a concentrator’s shift team a reliable reading of its ore before that ore reaches the mill, and to turn the reading into a plant setting that a person can approve. This report sets out what we built, how we tested it, what it would cost and earn, and how staged local validation would test it on South African ore. It describes a research prototype: the results are on public analogue data and a simulated plant. It is written for metallurgists, mineralogists, process engineers and the people who fund their technology. Technical terms are defined in the glossary (Appendix A).

1.1 Problem statement

The brief states the problem plainly: “without real-time feedback, processing plants cannot adjust to changes in ore quality, leading to inefficient chemical usage and lower mineral yields” [1]. Ore reaches the mill within minutes of leaving the belt, whereas assays and automated mineralogy are batch laboratory work. Mintek’s own Minerals Processing and Characterisation Cluster runs QEMSCAN automated mineralogy [2], and a commercial laboratory describes mineralogy as “not an overnight assay service” [3]. Consequently, each shift sets the mill and the flotation circuit for ore whose mineralogy it has not yet seen.

The cost of deciding blind falls most heavily on South Africa. Figure 1 illustrates world platinum-group metal (PGM) mine production in 2025, highlighting that South Africa mined an estimated 120 of the world’s 170 tonnes of platinum (about 71%) and 70 of 190 tonnes of palladium. It also holds 63,000 of the world’s more than 76,000 tonnes of

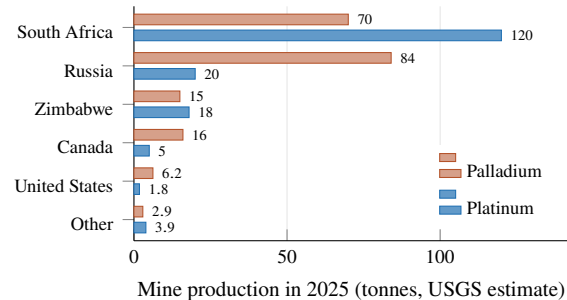


Figure 1: World platinum and palladium mine production in 2025. South Africa mined 120 of 170 tonnes of platinum; Zimbabwe’s Great Dyke ores are of the same family. Data: USGS [4].

PGM reserves, mostly in the Bushveld Complex [4]. The same source attributes the 9% fall in South African PGM output in 2025 to lower palladium prices, higher deep-level mining costs and disruptions to the electricity supply.

Recovery is where the money is. UG2 ore carries about 30% Cr_2O_3 against about 0.1% for Merensky ore, and the smelter caps the chromite in concentrate. Mintek testwork made UG2 concentrate at 87% recovery with 2.9% Cr_2O_3 , and recoveries above 90% were possible only if the Cr_2O_3 limit was relaxed [5]. Producers report recovery moves of this size to investors: Valterra raised concentrator recoveries by 1.2–1.7 percentage points at Mototolo and Amandelbult in 2025 [6]. Comminution, which sets both liberation and mill throughput, used 52% of site electricity in a benchmark of mines [7]. Plants do not decide entirely blind: operators, advanced process control (APC) and the mine plan already carry information about the ore. Thus, the decision we aim to improve is narrower: setting mill feed for each parcel of ore before it arrives, with a stated

Table 1: The brief and the judging criteria: evidence and status. Demonstrated = shown on public analogue data; partial = part shown; proposed = future work.

Requirement	Evidence (section)	Status
≥ 3 mineral phases	Pyrrhotite, chalcopyrite, pentlandite on analogue micrographs (4.4)	Demonstrated
Accuracy report	Belt R^2 , bound coverage, baselines; microscope per-class and decision-level results (4.1–4.5)	Demonstrated
Adjust a plant parameter	Feed-rate proposal (designed); regrind advice to a simulator over OPC UA (2.3)	Partial
Real time	Hardness: 92 ms per parcel. Mineralogy: laboratory cadence, not on the belt	Partial
Local validation	South African ore, independent test sets (8)	Proposed
Innovation	Bounded hardness input that abstains, with operator review (2)	Demonstrated
Feasibility	Staged plan, costs, support, compliance, risks (6–8)	Proposed
Impact	Scenario value per 1% more metal (6.4)	Scenario
Technical execution	Out-of-fold tests, baselines, failed ideas reported; 516 software tests (4)	Demonstrated

uncertainty, at a cost a producer can recover.

1.2 Our solution: three speeds, one decision

REEFPRINT/KHANYA reads the ore at three speeds (Section 3):

1. **the belt** (REEFPRINT), every 30 seconds: a short-wave infrared (SWIR) camera above the mill-feed conveyor feeds a model that predicts each parcel’s grinding hardness with an upper bound, and Bond’s law turns the bound into a mill feed-rate proposal. The belt layer estimates hardness, not mineral identity;
2. **the laboratory** (KHANYA), once a shift: a microscope model classifies sulphide minerals in polished sections, which addresses the brief’s three-phase requirement on slower, prepared samples;
3. **the decision**: an operator reviews each proposed change before the ore reaches the mill; the site’s existing control system keeps authority, and every step is signed and recorded.

It is important to note what is not claimed: real-time mineralogy on the belt, performance on South African ore, a measured plant benefit, or identification of chromite and the silicates. Table 1 maps the brief and the judging criteria published in the hackathon FAQ [8] to the evidence, with the status of each.

2 How the system works

2.1 Data

No South African belt scans or labelled Bushveld micrographs were available, so every model was trained and tested on public data (Table 2). HIDSAG supplies hyperspectral scans of Chilean porphyry copper–molybdenum drill core with laboratory Bond work-index values [9]. LumenStone S2, a Norilsk nickel–copper sulphide ore [10], shares the mineral assemblage of Bushveld ores but not their abundance: sulphides cover 62.8% of its sections by area, whereas Merensky ore contains up to about 3% base-metal sulphides and UG2 less [5]. Nevertheless, it is the right teacher for the phases, because the

Table 2: Datasets. All are public; none was supplied by Mintek or a producer.

Dataset	Ore	Size and use	Licence
HIDSAG [9]	Porphyry Cu–Mo, Chile	146 drill-core composites with Bond work index; visible (39 bands) and SWIR (75 bands). Hardness model	CC0
LumenStone S2 [10]	Ni–Cu sulphide, Norilsk	49 micrographs: 37 train, 12 held-out test. Microscope phases	Research use
Bushveld assays [11]	Bushveld chromitites	1,112 intervals. Check on PGE routing	CC BY 4.0

base-metal sulphides in UG2 are predominantly pentlandite, chalcopyrite, pyrrhotite and pyrite [5].

2.2 The belt layer: hardness from the infrared

Sensor (design). The design uses a SWIR line scanner such as the Specim SX25 (960–2500 nm, 640 pixels across, up to 162 lines a second) with a visible–near-infrared (VNIR) companion camera [12]. On a 1.2 m belt moving at 2 m/s it records a line every 12.3 mm with 1.88 mm pixels. It therefore measures parcels of ore, not individual particles. In the pilot design a *parcel* is 30 s of belt (about 60 m, or 4,860 scan lines averaged into one spectrum); in the replay reported here, a parcel is one HIDSAG composite scan of 80×117 pixels.

Model. Each parcel’s spectrum is reduced to features from the infrared absorption regions of clay, mica, chlorite and carbonate minerals (Fig. 4), and a model predicts the Bond work index (W_i , kWh/t). Results are reported only for samples the model never saw (out of fold), and a result counted only if it beat simple baselines under pass marks committed to the repository before each run. Five targets and several model variants were examined and only the work index survived, so the result is exploratory selection, not a confirmatory test; HIDSAG has no drill-hole identifiers, so grouped independence of the folds cannot be verified.

Bound and gate. The plant setting uses an upper bound that the true hardness should stay below 9 times out of 10 (a one-sided 90% split-conformal bound [13]), not the best guess. A more data-efficient bound [14] was tried first and failed its pass mark, so the exact bound that the plan named as its fallback is used, and the result is provisional. The 90% is a marginal coverage target that assumes new parcels resemble the calibration data; it is not a per-parcel guarantee and does not certify protection against overload. An out-of-distribution (OOD) check flags parcels unlike anything in training, and their advice is withdrawn.

2.3 The decision layer and its cadence

The decision layer turns the hardness bound into a mill setting that a person can approve. Ball mills are usually limited by motor power, so the tonnes a mill can grind depend on how hard the ore is. Bond’s law gives the specific grinding energy [15]

$$W = 10 W_i \left(\frac{1}{\sqrt{P_{80}}} - \frac{1}{\sqrt{F_{80}}} \right) \text{ kWh/t}, \quad (1)$$

where F_{80} and P_{80} are the feed and product sizes (μm) passed by 80% of the material. At a fixed grind target the feed rate

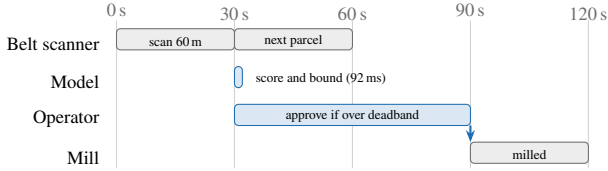


Figure 2: Proposed decision cadence for the pilot, to be agreed with the site. A parcel is scored as soon as it is scanned; the operator then has the rest of the assumed 90 s belt-to-mill transit to approve a change. Once a shift, the metallurgist approves the envelope, ramp limits and deadband. No approval in time, or a flagged parcel, means the advice is withdrawn and the site’s control system continues.

the mill can carry is proportional to $1/W_i$, so the proposal is

$$F_{\text{proposed}} = F_{\text{design}} \times \frac{W_{i,\text{design}}}{W_{i,\text{bound}}}. \quad (2)$$

The ratio removes explicit power and size terms only when power, feed size, product target and circuit efficiency are held constant; the pilot must test that approximation for the selected circuit. For example, a design hardness of 16.6 kWh/t and a bound of 16.9 kWh/t give 98.2% of design feed. The proposal must sit inside an operating envelope set by the site’s metallurgist; the demonstration envelope is 85–110% of design feed. The demonstration falls back to 100% of design feed; at a site, the fallback is the host’s own operating policy under its existing controls.

Figure 2 illustrates the proposed cadence, highlighting that a person approves changes, not every computation. A new proposal is computed every 30 s, but the operator sees it only when it differs from the running setpoint by more than a deadband (for example 2% of design feed) set by the metallurgist; smaller changes are not made. The operator may approve, modify within the envelope, reject or escalate. If nobody approves before the parcel reaches the mill (an assumed 90 s from the end of the scan), or the parcel is flagged, or the data are stale, the advice is withdrawn and the site’s own control policy continues; the research system never raises feed on its own. Feed may be cut by up to 10% but raised by only 3% per parcel. At most 120 proposals an hour are computed, but only changes larger than the deadband reach the operator, so far fewer are expected. The replay data have no time order, so the shadow phase measures the actual operator-facing rate before anyone acts on it. Every event is written to a hash-chained record with conventional (Ed25519) and post-quantum (ML-DSA-65 [16]) signatures; this supports traceability, not the correctness of the measurement. The microscope’s advice reaches a simulated regrind setting over OPC UA [17] (simulated); the feed-rate path is designed, not built.

2.4 The laboratory layer: minerals under the microscope

The microscope model is a standard image-segmentation network (DeepLabV3 with a ResNet-50 backbone [18, 19]) that labels every pixel of a reflected-light image as resin, pyrrhotite, chalcopyrite, pentlandite or magnetite. Only the approved checkpoint (SHA-256 de7135a9, full hash in Appendix D) may send advice to the simulator. A *Quick* scan classifies six 512-pixel fields of a section; a *full* scan covers the whole section.

Connected regions of the mask are split into grains. For

this demonstration, chalcopyrite and pentlandite are treated as payload and pyrrhotite as reject; that assignment is a demonstration assumption for this ore, because pyrrhotite’s value depends on PGM deportment and the host’s treatment objectives. A grain whose area is at least half payload is *payload-dominant* (shown as “free” in the application), otherwise *reject-dominant* (“locked”). This is a 2D payload-area proxy, not a measure of liberation or recoverability; the 50% line is a declared demonstration choice. A rule-based advisor turns these measures into *grind finer*, *continue*, *verify before acting* or, when the rejected phase dominates, *adjust reagent*. Each is a hypothesis for a metallurgist to test, not an instruction. It answers *verify* when the payload-area estimate lies within ± 0.335 of the 50% line, and gives no recommendation with fewer than 9 payload-bearing grains or mean confidence below 0.85. The ± 0.335 band was calibrated on the 12 evaluation sections (Section 4.5); the other thresholds are placeholders for site metallurgists to set.

2.5 How we tested it

Belt models were scored out of fold against the strongest simple baseline (the fold mean, the fold median and a colour-only model), with pass marks committed to the repository before each run. The microscope was scored on the publisher’s 12 S2 test sections. They were excluded from network training but reused during development, to compare pipelines, select the checkpoint and calibrate the advisor band, so we report them as *development-influenced evaluation sections*, not an independent test. An independent, source-disjoint local test set is the first deliverable of the laboratory study. Accuracy is the intersection over union (IoU: the overlap between predicted and expert-labelled areas, from 0 to 1), pooled over 103.8 million pixels and averaged per section, with bootstrap confidence intervals over sections. Pixels are correlated within a section, so sections, not pixels, are the independent units. Decision-level agreement runs the same advisor on the model’s mask and on the expert’s mask; a disagreement is *unsafe* if the model gives a confident instruction that the expert mask does not support. This measures sensitivity to segmentation error, not whether the rules themselves are metallurgically right.

3 System architecture

Figure 3 illustrates the system, highlighting that both layers pass the same gates and the same operator review. The belt and laboratory layers run on different time scales: laboratory results give slower context and audit evidence, not a check on each 30 s parcel. The belt monitor (REEFPRIENT Live) runs the hardness model, the decision view and the signed record. The microscope workbench (KHANYA) is a web application (React) on a Python service (FastAPI and PyTorch) that runs on an ordinary laptop processor. In local mode it accepts connections only from the same machine and makes no external requests, so a network outage at the plant does not stop the advice. No language model computes a measured value or a control value.

Speeds were measured on a four-core laptop processor: 92 ms per parcel (median); 16.7–21.2 s per six-field microscope scan on mains power (5 runs) and 30.8 s on battery. A static public demonstration runs the belt monitor on pre-computed data; a secure server passed 17 of 17 security tests

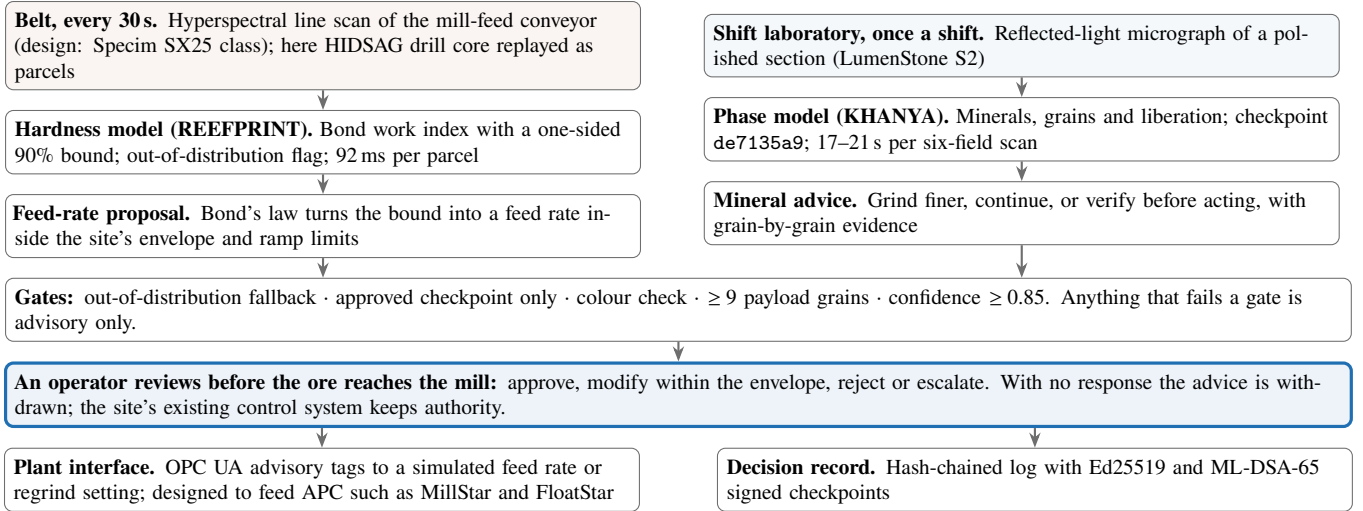


Figure 3: System architecture: three speeds, one decision. The belt layer (left) is the site system; the microscope layer (right) is the laboratory truth layer and delivers the brief’s three phases. Both pass the same gates and the same operator review. Built: the models, applications and signed record (on replayed public data). Simulated: the plant and regrind path. Designed: the belt sensor and feed-rate path.

locally and is not deployed. Software verification is in Appendix D.

4 Results

4.1 Belt: predicting grinding hardness

Of the five HIDSAG targets tested (copper and molybdenum recovery, lime consumption, pH and Bond work index), only the work index beat its strongest baseline under every pass mark. Figure 5 illustrates the out-of-fold results, highlighting that the infrared carries most of the information: the hyperspectral model explains 48% of the variance in work index, a model given only colour explains 21%, and the average of the training data explains nothing.

Figure 6 illustrates how the bound behaves, highlighting that it was exceeded on 10 of 146 samples (6.8%). The 95% interval for coverage (88.4–96.6%) spans the 90% target, so the result is compatible with it but does not establish future coverage, particularly on new ore. The price of the bound is width: the bound sits on average 1.64 kWh/t, about 11% of the mean hardness, above the best guess. The OOD check passed 137 parcels, marked 5 borderline and refused 4. Figure 4 shows one parcel as the model sees it. For parcel GMET-0004 the predicted W_i is 15.9 kWh/t and the bound 16.9 kWh/t, against a design hardness of 16.6 kWh/t (the training data’s 90th percentile), so the proposal is 98.2% of design feed. The laboratory value was 12.1 kWh/t, so the bound was conservative here.

4.2 Belt: from hardness to a plant decision

Figure 7 illustrates four feed-rate policies replayed over all 146 composites against a fixed design-feed policy (the 90th-percentile hardness), highlighting the trade-off between throughput and overloads. In the replay a parcel is *overloaded* when its true hardness exceeds the hardness the feed was set for; this is a model quantity, not a plant trip. Setting the feed from the bound, with OOD parcels withdrawn, gave a simulated throughput gain of +1.9% (95% CI +0.9 to +2.9%), and the overload share was 6.8% against 8.9%; the one-sided 95%

upper bound on the difference (+1.4 points) is inside the +3-point margin in the evaluation plan. The original replay did not apply the 85–110% envelope; a later analysis with it gives +1.7% (+0.7 to +2.7%) and the same overload share. Feeding from the best guess instead would add 13.6% but overload 45.2% of parcels, which is why the policy uses a bound. Perfect knowledge would add 14.0%, so the bound captures about one-seventh of what is available: its width (1.64 kWh/t) is where the rest is lost, and a tighter bound from site data is the main route to a larger gain.

The result is provisional for three reasons: the exact bound was adopted after the more efficient one failed, HIDSAG has no time order (so transit time, stockpile mixing and the asymmetric ramp limit are not simulated), and the ore is not Bushveld ore. Symmetric ramp limits were tested over 200 random parcel orders: $\pm 10\%$ per parcel left overloads unchanged (−0.3 points), but $\pm 5\%$ added 2.1 points and $\pm 2\%$ added 4.7, so feed must be cut fast. A robustness test defined before it was run found that a new capture of the same ore keeps its error ($\times 1.00$), as do half the field of view, added noise and $\pm 15\%$ illumination; a wavelength drift raised the error by a factor of 1.25 unflagged, so a wavelength-calibration check was added to the design.

4.3 Is it physically possible?

Every claim was tested against its physics, and the failures stay in the report. Bond’s law gives the throughput gain above under its stated assumptions (provisional). In the evaluated flotation scenarios, grinding faster cost up to 0.25 percentage points of recovery on the steep part of the recovery curve and none on the plateau; this is the largest loss among the scenarios modelled, not a physical bound, and the site’s recovery curve must be measured. Re-ordering ore during load-shedding cannot create tonnes, because energy per tonne is conserved over a cycle: it gave −3.8% in simulation, and even perfect knowledge gave −3.0%, so that claim was rejected. The camera’s 12.3 mm lines and 1.88 mm pixels resolve parcels, not particles; no test showed that the belt camera can identify minerals, so the microscope carries the phases; and the system is optical only, with no radioactive source and no radiation sensing.

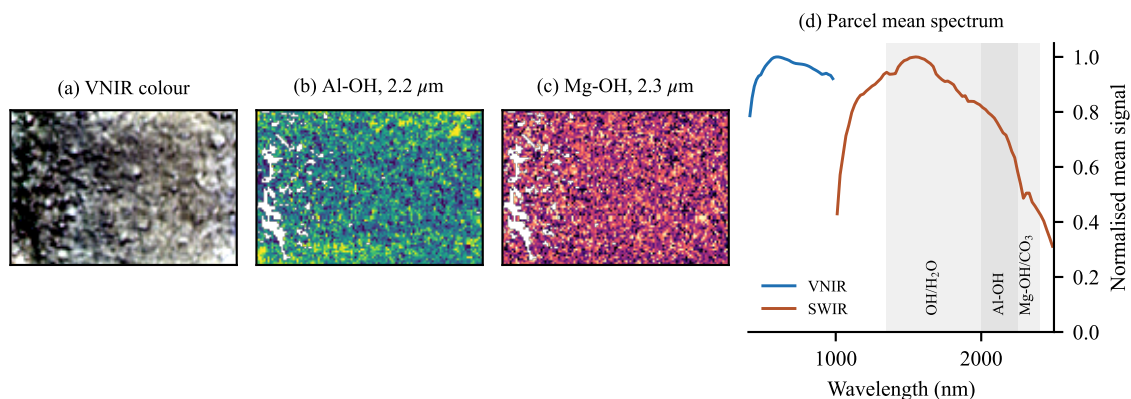


Figure 4: One belt parcel as the hardness model sees it: HIDSAG composite GMET-0004 (Chilean porphyry Cu–Mo), 80×117 pixels. (a) Colour image from the visible bands. (b, c) Relative absorption depth at the Al–OH (white mica, clays) and Mg–OH (chlorite, biotite, carbonate) features; detector stripes removed for display only. (d) Mean spectrum, with the absorption regions the model uses. The model predicts W_i 15.9 kWh/t with a 90% upper bound of 16.9 kWh/t (laboratory value 12.1 kWh/t). The maps show absorption features, not a mineral identification.

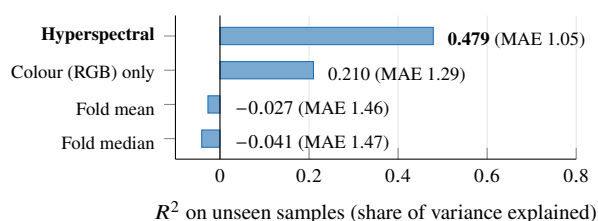


Figure 5: Bond work index on 146 HIDSAG composites, out of fold; mean absolute error (MAE) in kWh/t. Mean measured W_i is 14.62 kWh/t.

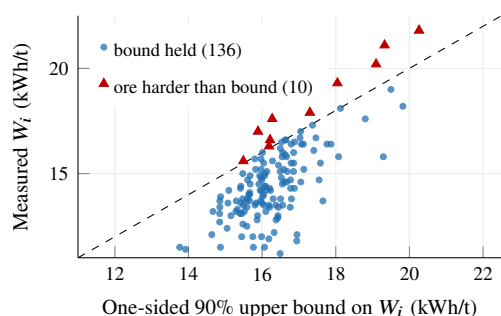


Figure 6: Calibration of the hardness bound, out of fold, 146 composites (no drill-hole grouping available). Points below the dashed line are samples the bound covered: 93.2% (95% CI 88.4–96.6%) against the 90% target. The bound sits on average 1.64 kWh/t (range 1.04–2.14) above the prediction.

4.4 Laboratory: phase identification

The microscope addresses the brief’s three-phase requirement on analogue ore. Figure 8 illustrates its accuracy by class, highlighting IoU values of 0.87 for pyrrhotite, 0.58 for chalcopyrite and 0.55 for pentlandite (precision and recall in Appendix B). The pooled mean IoU is 0.5725 (95% CI 0.494–0.624); averaged per section it is 0.4671 (0.412–0.534). The model clearly beats trivial baselines (majority class 0.117; colour-only 0.395).

Magnetite, a fourth class the brief does not require, is never predicted. It is only 0.8% of the test pixels: 78% of it is labelled as resin, 17% as pyrrhotite and 5% as pentlandite. The last share is about 0.3% of all pixels predicted as pentlandite, so magnetite barely moves the payload measures; where it is intergrown with a sulphide grain, dropping it slightly over-

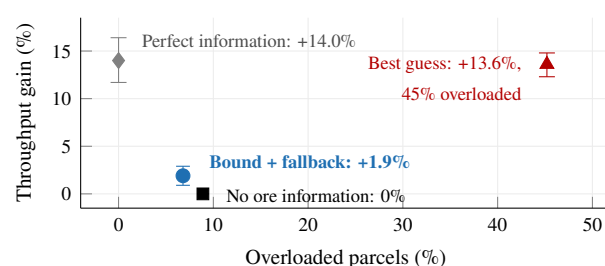


Figure 7: Feed-rate policies replayed over 146 composites against a fixed design-feed policy (simulated; not PGM ore, not a time series, original replay without the envelope), with 95% confidence intervals. Up is more throughput; left is fewer replay overloads.

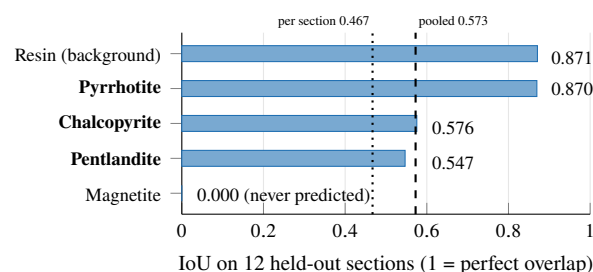


Figure 8: Microscope accuracy by class for the approved checkpoint, full-section inference. The three phases the brief requires are in bold. Section means range from 0.340 to 0.728 (Appendix B).

states that grain’s payload share.

A retrained candidate (42646cfa) scores 0.632 overall and does detect magnetite (IoU 0.25), mainly because chalcopyrite improves (0.800 against 0.576). However, it loses pyrrhotite (0.818 against 0.870) and pentlandite (0.446 against 0.547), and only half of the pixels it calls pentlandite are pentlandite (precision 0.50 against 0.68). Pentlandite is the main nickel and PGM carrier, and false pentlandite would inflate the payload share. The approved checkpoint was therefore retained on a precautionary rationale; the comparative decision safety of the two models has not been tested, and a second training run of the approved recipe scored 0.454 because the patch sampler was not seeded; the sampler is now seeded (Appendix D).

Table 3: Decision-level confusion on the 12 held-out sections: advice from the expert mask (rows) against advice from the model’s mask (columns). The advisor can also flag a reagent issue when the reject phase dominates; this happened once, on the expert mask only.

Expert mask \ model	Grind	Continue	Verify	Too few	Total
Grind finer	3	0	3	2	8
Continue	0	1	2	0	3
Adjust reagent	0	0	1	0	1
Total	3	1	6	2	12

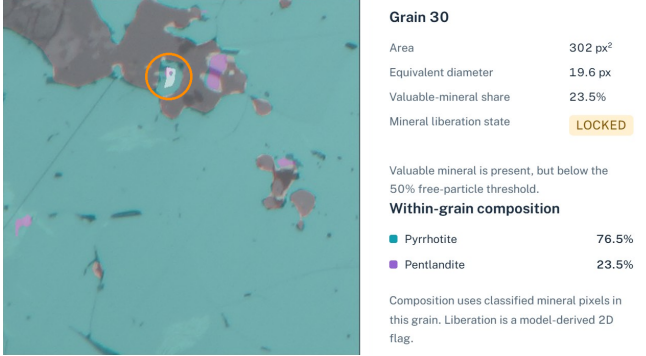


Figure 9: The grain inspector on section test_11 (Quick scan): one analysed field with the phase overlay (left; teal pyrrhotite, violet pentlandite) and the selected grain’s evidence (right). Grain 30 (circled) is 76.5% pyrrhotite and 23.5% pentlandite, so the application labels it locked (reject-dominant on the 2D proxy); grain 31, to its right, is pure pentlandite. Cropped from a screenshot of the application.

4.5 Laboratory: decisions and grain evidence

Table 3 compares the advice from the model’s masks with the advice from the expert masks. The model produced 4 definite recommendations, 6 *verify* outputs and 2 *too few grains* outputs on the 12 sections, so its action coverage was 4 of 12 (33%). No definite recommendation disagreed in the unsafe direction with the same advisor applied to expert masks. This is a small, development-influenced rule-agreement result: it does not validate the rules themselves, establish an independent unsafe-action rate or demonstrate plant safety. The one-sided 95% upper bound of 22.1% on 0 of 12 refers to all sections, not to risk when the system acts. Both the rules and the abstention thresholds need independent evaluation on the host stream.

It is important to note that the ± 0.335 verify band was calibrated on these same sections, so the agreement result is partly in-sample. Replaying the liberation rule on the stored section measurements shows how much it depends on the band: on the 9 sections it governs, the result stays at 0 wrong for bands of ± 0.25 or wider (7 correct at ± 0.25), but one section (test_04) becomes unsafe below that, and two with no band at all. The band must therefore be re-set on independent sections in the laboratory pilot. A realistic exposure shift (-35 levels on each colour channel) changed the advice on 8 of 12 sections, so illumination control is also a pilot requirement.

Figure 9 illustrates the grain inspector, highlighting how a mineralogist can audit the advice grain by grain. In Quick mode each field is classified without context beyond its edge; on this one example, test_11, 19 of 24 grains come within 6 pixels of a field edge, which shows a potential measurement problem rather than a dataset-wide error rate. Thus, Quick-mode grain statistics are indicative; a border policy (excluding or reconciling edge grains) must be set and checked against a

reference analysis.

5 Comparison with literature and limitations

5.1 Comparison with literature

Published work supports the plausibility of the approach, not its validity here. Notole et al. [20] validated hyperspectral imaging and portable XRF against reference methods on Merensky Reef drill core scanned at the Council for Geoscience in Pretoria; that shows the sensing works on Bushveld material, not that our hardness model does. The three sulphides the microscope classifies are the main base-metal sulphides of UG2 ore [5], and our 50% proxy line coincides with a boundary between middling classes in Mintek’s liberation scheme [2], although that boundary does not by itself define free material. Mintek’s AI optimisation raised a lead–zinc concentrator’s throughput by 0.5%, worth over US\$2 million (about R33 million) a year [21], which shows that moves of this order are valued, not that this system would deliver them.

Four ideas were tested and withdrawn, and they mark where the method stops. The belt spectra did not predict QEMSCAN mineralogy once size fraction and process line were accounted for. Belt chemistry routed Bushveld PGE ore worse than the mine plan’s seam label (balanced accuracy 0.79 against 0.82). The belt did not predict recovery, lime or pH better than a baseline. Plant tags never forecast concentrate silica an hour ahead better than the last laboratory assay.

5.2 Limitations

The following limitations apply, and (a), (b) and (e) block any investment decision until tested: (a) no South African ore has been evaluated, and a model trained on Chilean porphyry core may not transfer, because hardness also depends on silicification, grain size and fracturing; (b) the belt result is a replay of 146 composites without time order or drill-hole grouping, against a fixed design-feed policy rather than the plant’s existing control, and a camera sees the belt surface, which may not represent the mass entering the mill; (c) the belt camera predicts hardness and does not identify minerals, and the microscope has not been trained on chromite or the silicates; (d) the 12 microscope sections guided development, and the approved checkpoint cannot be regenerated exactly; (e) the advisor’s rules (including the pyrrhotite-reject and 50% assumptions) are unvalidated for any host ore, its band was set on the evaluation sections, and liberation is a 2D proxy; and (f) latency is measured on one laptop, the plant is a simulator, and economic benefit has not been attributed at any site. Compared with the abstract Mintek accepted, three things changed: the five Bushveld phases became three sulphides on an analogue ore, because no labelled Bushveld images existed; the QEMSCAN “teacher” became the pilot’s truth-laboratory stream; and the fallback was reversed from holding the last setpoint to applying the conservative setting.

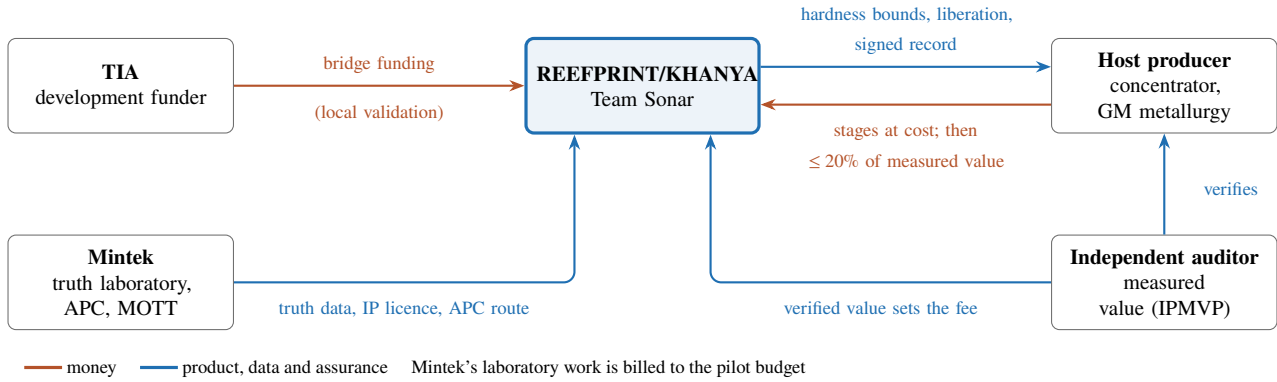


Figure 10: The proposed business model; partner roles are proposals, not agreements. The producer would pay for development stages at cost; after a successful trial it would pay a capped share of independently measured value, so no measured value means no fee. Mintek would earn from the laboratory work the models need; licensing follows the hackathon IP agreement with its Office of Technology Transfer (MOTT).

6 Business model and financial case

6.1 Value proposition and business model

Our value proposition is simple: the plant gets a reading of its ore before the mill, with a stated bound, and after the trial it pays only for value that an independent party has measured. No producer, Mintek or TIA has agreed to fund, host or license the work; the roles below are proposals. The buyer is the concentrator manager or general manager for metallurgy at a South African PGM producer, who owns recovery, concentrate quality and mill throughput; producers such as Valterra, Implats, Sibanye-Stillwater, Northam and Tharisa are market segments, not customers. Figure 10 illustrates how money and value flow, highlighting the two kinds of payment: the pilot is paid at cost, and only the later fee depends on measured value. The Technology Innovation Agency (TIA), the national agency that funds technology development towards commercialisation [22], is a possible funder of the local validation stage, subject to its programme criteria.

The revenue model has four streams: a paid laboratory study; a monthly licence for the laboratory software; recovery of shadow-sensing costs; and, after a successful trial, at most 20% of measured value per concentrator per year. The model fits Mintek’s published work [21]: the models need QEMSCAN composites and Bond tests, which is laboratory work; a hardness input could add ore-side information to process control that today works from process data; Mintek’s benefit-sharing model with Hindustan Zinc has generated over R35 million since 2021; and its products reach over 60 countries through distributors such as Molycop and Mercantil. South African PGM output also fell 9% in 2025 [4], while Valterra’s realised basket price rose from R32,611 per ounce in 2025 to R45,993 in the first half of 2026 [6, 23].

Table 4 compares the options a plant already has. The system replaces none of them; whether it adds information beyond existing APC, operator practice and mine-plan blending is what the local trial must show.

6.2 Pricing, and what the money pays for

Table 5 sets out a staged plan in which the cheapest experiment that can settle the largest uncertainty comes first, and what each stage pays for beyond equipment and software. Stage A tests the microscope and the spectral-hardness relationship on the host’s own ore before any belt equipment is bought. Sepa-

Table 4: What plants use today, and where this system differs.

Option	What it gives	Where this differs
Cross-belt neutron analysers	Bulk elements, minute by minute	Hardness, not elements; isotope sources need a SAHPRA licence [24], this needs none
Slurry analysers	Grade and Cr_2O_3 in slurry, after milling	Reads the ore before the mill; their output is an input
Automated mineralogy (QEMSCAN)	Phases and liberation, in days	Predicts between truth results; its composites retrain the models
APC (e.g. MillStar, FloatStar)	Stable, optimised circuits from process data	A partner: an ore-side input inside APC limits

rate quotes are needed for each work package.

6.3 From throughput to recovered metal

The belt result is a throughput gain, and revenue comes from recovered metal, so the two must be bridged. For a consistent stream, metal output is proportional to dry tonnes $\times$ head grade $\times$ recovery, so

$$\frac{\Delta M}{M} = \left(1 + \frac{\Delta T}{T}\right) \left(1 + \frac{\Delta G}{G}\right) \left(1 + \frac{\Delta R}{R}\right) - 1, \quad (3)$$

where T is throughput, G head grade and R recovery. With unchanged grade, a 1.75% throughput gain and a recovery loss of 0.25 percentage points from 87% (a relative change of -0.29%) give 1.46% more metal. Where the plant is limited by ore supply, extra throughput is worth nothing. If all of Zondereinde’s output passed the instrumented circuit (an upper bound), 1.46% more metal would be worth R64–123 million a year in margin, and a quarter of that gain R16–31 million; these are simulated gains, not measurements.

6.4 Financial methodology

The following assumptions apply: (a) annual output is that of Northam’s Zondereinde mine, 333,050 equivalent refined 4E ounces in F2026 [26]; (b) prices are Valterra’s realised basket prices of R32,611 per ounce (2025) and R45,993 (first half of 2026), with cash operating costs of R19,488 and R20,677 per ounce [6, 23], used as proxies for another company’s mine, with different baskets, payability and cost structure; (c) costs are the ranges in Table 5; and (d) the equipment has a 5-year

Table 5: Staged costs. Prices are the team’s hypotheses (H) or planning assumptions (A) until vendors and laboratories quote; ranges first set in US\$ are converted at R16.4 per US\$, the January–September 2026 average [25]. Each stage starts only if the previous gate passes.

Stage	Price	What it pays for	Who could pay	Basis
A1. Laboratory study (KHANYA), about 3 months	About R210,000	Team time to retrain and freeze the microscope on QEMSCAN-labelled sections of a named host stream; an independent test set; a validation report; laboratory training	Development funder (e.g. TIA)	H
A2. Hardness feasibility, about 4 months	To be quoted	Paired host samples: Bond tests at an accredited laboratory and hyperspectral scans on an existing or shared scanner; a frozen local model evaluated on held-out samples	Development funder; host in kind	A
B. Shadow sensing, one belt	R8.2–49.2 million installed; R1.6–9.8 million a year to run	<i>Installed:</i> cameras, enclosure, lighting, references, encoder, edge computer, structural and electrical work, historian integration, OT security assessment. A leased or temporary system is an option to quote. <i>Running:</i> sampling, assays, Bond tests, QEMSCAN composites, maintenance, support, training	Host producer, at cost and at risk	A
C. Advisory trial	Running costs of B, plus independent measurement and verification	Operator protocol, on/off operating blocks, metal accounting	Host producer	A
Contingency	20% of installed cost (R1.6–9.8 million)	Released only against quotes	Funder of each stage	A
After a successful trial	At most 20% of measured value (fee basis to agree)	Models, audits, support; zero if no value is measured	Producer	H

life at a 10% discount rate; and (e) an affected-output fraction f of the mine’s output passes the instrumented circuit, taken as 1 unless stated. The annual value of an uplift u in recovered metal is

$$V = u \times f \times Q \times m, \quad (4)$$

where Q is annual output in ounces and m the margin per ounce. For $u = 1\%$ and $m = \text{price less cash cost}$,

$$V = 0.01 \times 333,050 \times (\text{R}13,123 \text{ to } \text{R}25,316), \quad (5)$$

which is equivalent to R43.7–84.3 million a year (R108.6–153.2 million in revenue before costs). Deducting the full average cash cost may be conservative where the mill has spare capacity, but extra mining, handling or bottleneck costs could change that, so a host-approved incremental cash-flow model is needed before investment. The installed cost is spread over the equipment’s life with the capital recovery factor

$$\text{CRF} = \frac{r(1+r)^n}{(1+r)^n - 1} = 0.2638 \quad (r = 10\%, n = 5), \quad (6)$$

so the annualised cost is

$$C = \text{CRF} \times C_{\text{installed}} + C_{\text{running}}, \quad (7)$$

which is equivalent to R3.8 million a year (low), R11.4 million (middle) and R22.8 million (high), or R25.4 million for the high case with contingency. The break-even uplift is then $u_{\text{BE}} = C/(f \times Q \times m)$. Table 6 shows how it moves with cost, margin, affected fraction and the fee basis. If the 20% fee is charged on the margin before system costs, the producer breaks even only when $0.8V = C$. Figure 11 illustrates the all-output case.

Annualising over five years describes continued successful use; pilot-stage cash exposure is separate. In the staged plan, a failure at the laboratory or hardness gates costs only Stage A (about R210,000 plus the quoted hardness work), because no belt equipment is bought before the hardness gate. Once Stage B starts, the producer’s committed exposure is the installed cost plus running costs to date: R8.5–50.8 million two months after installation and R9.4–56.6 million after nine, or up to R66.4 million with the full contingency, before any recoverable equipment value. The fee basis (before or after system costs) must be agreed, so that no support cost is charged

Table 6: Break-even sensitivities (recalculated under the stated assumptions; illustrative, not a site funding threshold).

Scenario	More metal needed
Low cost, 2026 margin, all output affected	0.045%
High cost, 2025 margin, all output	0.52%
High cost + contingency, 2025 margin, all output	0.58%
High cost, 2025 margin, 20% fee before system costs	0.65%
High cost, 2025 margin, half of output affected	1.04%
High cost, 2025 margin, a quarter of output affected	2.09%
High cost, 30% lower basket price (2025 costs)	2.05%

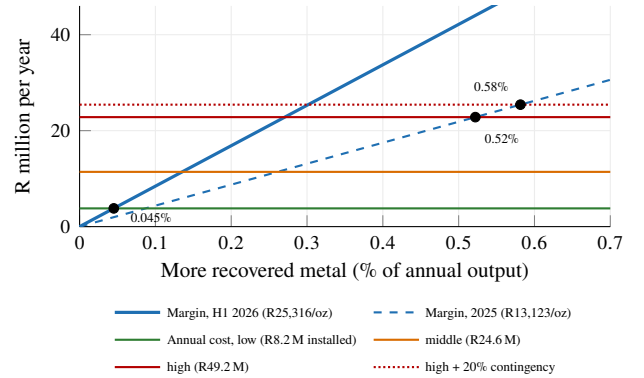


Figure 11: Break-even if all of Zondereinde’s output were affected ($f = 1$): annual margin on extra recovered metal (blue) against the annualised system cost (horizontal lines), before any fee.

twice through the licence, running budget and fee. Measured value requires an agreed metallurgical accounting boundary and attribution method under IPMVP [27].

6.5 Risks, contingency and fallbacks

Table 7 sets out the main risks of using the product, as the team assesses them. Each has a mitigation that acts before it happens and a fallback that limits the damage if it does. Three fallbacks run through all of them: the plant’s existing controls keep authority; each stage ends at a gate where spending can stop; and the fee is zero if no value is measured. Fallbacks reduce exposure to detected failures; they do not guarantee safety against undetected ones.

Table 7: Risk register: mitigation before the event, and the contingency or fallback if it happens.

#	Risk	Mitigation (before)	Contingency or fallback (if it happens)
R1	The hardness model does not transfer to Bushveld ore	Hardness feasibility on host samples (H1) before any belt purchase	Stop after Stage A; no belt spend
R2	Belt conditions: dust, moisture, vibration, lamp ageing, wavelength drift	Purge air, calibration references and a calibration gate at commissioning (S1)	Detected faults withdraw advice; undetected drift remains a risk, checked by monthly audits
R3	Power outages and load-shedding	Runs offline on an edge computer with a UPS	After a restart, advice is blocked until checks pass
R4	Cyber attack or an unsafe write to the plant	Read-only advisory tags; no write path to safety systems; IEC 62443 zones; signed records	Disconnect; the plant continues on its existing control system
R5	Operators ignore, over-trust or cannot keep up with the advice	Deadband and shift-level approvals; training before the advisory phase; named roles	Advice expires; site control continues
R6	No measurable gain	On/off operating blocks with confidence intervals; independent measurement and verification	Stop at V1; no fee is due
R7	Cost overrun or a weaker rand	Quotes before Stage B; 20% contingency; funding by gate	Defer Stage B; continue the laboratory work
R8	A fall in PGM prices	The fee is a share of measured net value	The fee falls with value automatically
R9	Funding delay, or staff not available (two of three team roles are filled)	Stage A is small and can start first; third role recruited or sourced in kind	Laboratory work only until funded and staffed
R10	Wrong advisor rules, or rights to use the training data commercially	Rules validated by a mineralogist on host samples; LumenStone is licensed for research use, so commercial rights are clarified with the rights holder	Display measurements without prescriptive advice; retrain on host-owned data

7 Implementation and support

The pilot depends on people more than on hardware. Appendix C gives the detailed plans; this section summarises them.

Support (Table 11). A host site would provide a sponsor, a process metallurgist, an analytical chemist, a process control engineer, an instrumentation technician and an OT security officer; Mintek a mineralogist and the IP route; Team Sonar three roles, two filled by the authors and one (software, security and deployment) to be recruited or sourced in kind.

Maintenance (Table 13). Calibration and data-quality checks withdraw advice when a specified failure is detected. They reduce exposure to known failure modes but cannot detect every harmful change; undetected drift, sampling mismatch and wrong rules remain trial risks.

Compliance (Table 12). These are planned controls, not achieved compliance. The Mine Health and Safety Act requires a risk assessment with the health and safety committee and training before significant changes to plant or equipment [28]; POPIA governs approvers' names in the record [29]; Mining Charter clause 2.3.1 counts skills and R&D spend [30]; IP from public funding falls under the IPR Act [31]; Mintek and TIA spend under the PFMA [32]; and the camera needs no radiation licence. Before installation, the health and safety committee, the management-of-change review, the IT/OT change board and the capital committee sign off.

Independent fact-checkers to outsource: (a) a SANAS-accredited laboratory for Bond tests and assays [33]; (b) Mintek's QEMSCAN for mineralogy; (c) an independent auditor for the value on which fees are charged, using IPMPV [27]; (d) an IEC 62443 assessor for operational-technology security [34]; (e) an IP attorney through MOTT; and (f) a university reviewer for the model audit.

Training (Table 14). Each group receives a short module before its phase starts, aligned with the site's workplace skills plan, which mines report to the Mining Qualifications Authority [35].

Table 8: Proposed gates. Thresholds marked "agreed" are set with the host before evaluation data are collected.

Gate	Evidence to pass	Decides
G0	Named host stream and bottleneck; sampling and data/IP access; quoted Stage A budget	Sponsor, funder
L1	Frozen microscope on an independent host test set meets agreed accuracy and bias; border policy checked	Mineralogist
H1	Frozen local hardness model beats baselines on held-out host samples, with agreed coverage <i>and</i> bound width	Process lead; no belt purchase before this
S1	Sensor availability, calibration, material timing, operator-facing proposal rate within agreed limits	Site operations
A1	Change and risk process passed; interface rights, expiry and fallback defined; measurement plan	Host
V1	Pre-specified trial: margin above break-even with uncertainty; site recovery curve and metal accounting reconciled	Pre-agreed investment rule

8 Roadmap: evidence before capital

Figure 12 illustrates the plan, highlighting that the transfer question is answered before any belt equipment is bought. The laboratory and hardness studies have separate gates: passing one is not evidence that the other has passed. Table 8 lists what each gate must show and who decides; the numerical thresholds are to be agreed with the host before data are collected, so that the study can return a clear negative result.

9 Conclusions and recommendation

The current work demonstrates an integrated research prototype with promising results on public analogue data. The hardness model improves on simple baselines, and the microscope classifies three relevant sulphide phases with uneven class performance. The feed-rate benefit remains a simulation, and the microscope decision result is development-influenced. The most consequential open questions are transfer to the host ore, the validity of the metallurgical rules, sampling and sensing representativeness, and attributable benefit. We therefore rec-



Figure 12: Proposed staged roadmap (planning estimate; dates depend on samples, test turnaround, procurement and site approvals). Shaded bars are the evidence-first stages; dashed bars are scenarios. Diamonds are go/no-go gates (Table 8). Readiness levels are awarded from demonstrated evidence, not elapsed time.

ommend a scoped local validation programme, a laboratory study and a hardness feasibility study on the host's ore, followed only conditionally by shadow sensing and a controlled advisory trial. Larger investment should depend on independently assessed evidence and a host-specific economic case.

CRedit authorship contribution statement. **Lethabo Hoaeane:** Conceptualisation, domain lead, belt hyperspectral modelling, belt monitor and decision application, economics, presentation. **Sibusiso Khumalo:** Microscope segmentation model, accuracy and decision-level evaluation, advisor and safety gates, OPC UA simulation, workbench review and verification, technical report.

Declaration of competing interest. The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Use of AI tools. Code and drafting were assisted by AI coding assistants. Every reported number was computed by code in the project repository and checked by the authors.

Data and code availability. HIDSAG is available under CC0 (doi:10.6084/m9.figshare.c.5983921) [9]; LumenStone under research-use terms [10]; the Bushveld assays under CC BY 4.0 (doi:10.17632/dc8jcnbcvk.1) [11]. Code, checkpoint hashes and every results file behind the numbers in this report: <https://github.com/Sibusiso-K/KHANYA> (Appendix D). A static demonstration of the belt monitor runs at <https://lethabomh14-reefprint.static.hf.space>.

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A Glossary

APC	Advanced process control, e.g. Mintek’s MillStar and FloatStar
Bond work index (W_i)	Energy needed to grind ore, in kWh/t; harder ore has a higher W_i
Bound	Here, a hardness the true value should stay below 9 times in 10 (one-sided 90% conformance bound)
Fallback	In the demonstration, 100% of design feed; at a site, the host’s own operating policy under its existing controls
Deadband	The smallest change worth showing the operator
Envelope	The range of feed rates the site’s metallurgist allows (demonstration: 85–110% of design)
Payload-dominant	A grain at least half payload by 2D area (“free” in the app); a proxy, not liberation
IoU	Intersection over union: overlap of predicted and expert areas, 0 to 1
IPMVP	International Performance Measurement and Verification Protocol
MOTT	Mintek’s Office of Technology Transfer
OOD	Out of distribution: unlike anything in training
OPC UA	The industrial data-exchange standard IEC 62541
Parcel	30 s of belt in the pilot design; one composite scan in the replay
PGM, 4E	Platinum-group metals; 4E = Pt, Pd, Rh and Au
QEMSCAN	Automated scanning-electron mineralogy
SWIR, VNIR	Short-wave and visible–near infrared light
TIA	Technology Innovation Agency
TRL	Technology readiness level, 1 to 9

B Microscope detail

Tables 9 and 10 (final page) give per-class results for the approved checkpoint and the retrained candidate, and the pixel confusion matrix.

Per-section mean IoU (test_01 to test_12): 0.413, 0.728, 0.659, 0.466, 0.472, 0.340, 0.453, 0.504, 0.375, 0.378, 0.377, 0.443. Chalcopyrite and pentlandite are confused with each other and with pyrrhotite; that, not magnetite, limits the payload measures.

C Implementation plans

Tables 11–14 (final page) give the support, compliance, maintenance and training plans.

D Software verification and reproducibility

Tests. The microscope workbench has 181 unit-test functions and 7 browser tests, with four continuous-integration checks passing; the belt software has 328 unit-test functions. Static security analysis reported no issues after a dependency upgrade.

Checkpoints (SHA-256). Approved:

de7135a96541a46dc0981a991cb954186c7cd669ea1c1b33914d1929ae9b1357.

Candidate:

42646cfafbeac5398386dba17f7ad3531ffcea572e6342cc4fcba42ed0b44b3b.

Reproducibility. The training patch sampler was unseeded, so a second run of the approved recipe scored 0.454; it is now seeded by seed, epoch and index, with a test that two runs draw identical patches (commit adeb1ee). Inference is reproducible from the archived approved weights; retraining is not: the approved checkpoint itself cannot be regenerated exactly, and seeded patch selection alone does not prove the full pipeline now reproduces it. The image-analysis runtime is pinned (requirements-lock.txt); pinning the training environment and a seeded re-run are pilot tasks.

Table 9: Microscope per-class results, approved checkpoint against the retrained candidate, 12 S2 test sections (full-section inference).

Class	Approved de7135a9			Candidate
	IoU	Recall	Precision	IoU (precision)
Resin	0.871	0.920	0.942	0.848 (0.960)
Chalcopryrite	0.576	0.716	0.746	0.800 (0.937)
Magnetite	0.000	0.000	–	0.248 (0.255)
Pyrrhotite	0.870	0.934	0.926	0.818 (0.950)
Pentlandite	0.547	0.742	0.675	0.446 (0.503)
Mean	0.573			0.632

Table 10: Pixel confusion for the approved checkpoint (% of each true class), 103.8 million test pixels.

True \ predicted	Resin	Ccp	Mag	Po	Pn
Resin	92.0	0.8	0.0	5.7	1.5
Chalcopryrite	1.6	71.6	0.0	3.1	23.7
Magnetite	78.3	0.1	0.0	16.8	4.8
Pyrrhotite	1.1	1.6	0.0	93.4	3.9
Pentlandite	0.7	0.8	0.0	24.3	74.2

Table 11: Support needed from the host site, Mintek and the funders. Time commitments are planning assumptions.

Role	Why the pilot needs them	When (stage)	Commitment
Plant manager or GM metallurgy	Sponsor; owns the budget, approves the envelope, decides at each gate	All	Gate meetings
Process metallurgist	Sets the envelope, deadband and fallback policy; reviews the record	A2–C	0.3 full-time
Mineralogist (Mintek)	QEMSCAN truth; labels sections for retraining and the independent test set	A1–C	0.2 full-time
Analytical chemist	Assays and Cr ₂ O ₃ ; quality control with blanks, duplicates and reference materials	A2–C	Per sampling plan
Process control engineer	OPC UA tags, historian access, later coupling to APC	B–C	As needed
Instrumentation engineer and technician	Installation, enclosure, references, lamps	B–C	0.3 full-time
OT and IT security officer	Network zone, read-only interface, backups	B	0.1 full-time
Health and safety representatives; information officer	Risk assessment before installation (MHSA); POPIA and the record policy	Before B	Consultation
Mintek MOTT; TIA portfolio manager	IP agreement; funding tranches at gates	G0, gates	As needed
Team Sonar	Models, software, integration, analysis and reporting	All	Two filled; one to recruit

Table 12: Laws, policies and standards that apply, and the planned controls (not achieved compliance).

Law, policy or standard	What it asks of this project	Planned control (evidence required)
Mine Health and Safety Act 29 of 1996 [28]	Assess risks of new equipment, consulting the health and safety committee (s. 11, s. 34); train employees before significant changes to plant or equipment, and keep records (s. 10)	Installed under the site's risk assessment and lockout; advisory only; operators trained before the advisory phase
Protection of Personal Information Act 4 of 2013 [29]	Lawful, limited processing of personal information, such as approvers' names	Purpose, access, retention and non-disciplinary use agreed with the information officer before the shadow phase
Mining Charter 2018, clause 2.3.1 [30]	Invest 5% of the leviable amount in essential skills development, including R&D in processing and technology efficiency	Pilot training and R&D may count toward it; clause 2.3 was not among the clauses set aside by the High Court in 2021 [36]
IPR from Publicly Financed R&D Act 51 of 2008 [31]	IP from publicly financed R&D is identified, protected and commercialised for South Africa (NIPMO)	IP managed by MOTT under the hackathon agreement, with invention credits [8]
Public Finance Management Act 1 of 1999 [32]	Mintek (Schedule 3B) and TIA (Schedule 3A) spend public money under PFMA controls	Funding released in tranches tied to gates G0–V1, with auditable deliverables
Hazardous Substances Act 15 of 1973 (SAHPRA) [24]	Licences for radioactive sources (Group IV)	Not applicable: optical sensing only
IEC 62443 [34]; OPC UA [17]	Zones and conduits for industrial cyber security; secure data exchange	Read-only advisory tags; no write path to safety systems; IT/OT sign-off
ISO/IEC 17025, accredited by SANAS [33]	Demonstrated competence of testing laboratories	Bond tests and assays used as truth come from accredited laboratories
IPMVP (EVO) [27]	A recognised protocol for measuring and verifying savings	Fees invoiced only on value measured under an agreed plan by an independent party

Table 13: Proposed maintenance plan; the site sets the final intervals.

Task	Interval	Owner
Reference checks; refuse if a feature shifts over half a band	Every reference cycle (automatic)	System
Window cleaning and purge-air check	Each shift, or on alarm	Instrument technician
Lamp replacement	Lamp-life schedule	Instrument technician
Enclosure temperature and dew point	Continuous alarm	System; electrical
Microscope colour and illumination check	Every image; daily	Laboratory technician
Model audit against Bond tests and QEMSCAN	Monthly in the pilot	Metallurgist, mineralogist
Retraining	Only through a validation gate	Team Sonar
Security patches; backup restore test	Monthly; quarterly	Team Sonar, IT/OT

Table 14: Proposed training plan. Records are kept as the MHSA requires.

Who	What they learn	Format; when
Control-room operators, supervisors	Reading a proposal; approve, modify, reject or escalate; the deadband; what the fallback does	Half a day plus supervised shifts; before C
Metallurgists	Setting the envelope and deadband; reading bounds and liberation; reviewing the record	One day; start of A2
Laboratory staff	Workbench, colour calibration, grain audit, comparison with QEMSCAN	One day; during A1
Instrument technicians	Camera, enclosure, references, lamps, lockout	One day with the vendor; start of B
IT and OT staff	Network zones, backups, patching, incident response	Half a day; before installation
Managers and funders	Reading the measured-value report and the gates	One hour; each gate